

Wave 3.1 Offset-Aware Probe Campaign Results

Wave 3.1 Campaign Closeout

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved `wave 3.1` offset-aware probe campaign. The executed package trained the first learned `sequential_residual_offset_probe` candidate across the three required direction surfaces: `global`, `Fw`, and `Bw`.

The campaign completed all 3 planned runs with zero training failures. A terminal-level `conda run` message appeared after the runner printed the completed-campaign summary, but the generated campaign artifacts show `0` failed runs, per-run process return code `0`, and no traceback in the run logs.

The scalar leaderboard ranks `Fw` first by `test_mae`, but that ranking is only a diagnostic ordering. `Wave 3.1` keeps three parallel best branches for future verification and deployment analysis: one `global`, one `Fw`, and one `Bw` candidate.

Campaign Artifacts

Artifact	Path
Campaign output	output/training_campaigns/2026-06-04-11-36-09_track2f_offset_aware_probe_campaign_2026_06_03
Campaign leaderboard	output/training_campaigns/2026-06-04-11-36-09_track2f_offset_aware_probe_campaign_2026_06_03/campaign_leaderboard.yaml
Branch scalar pointer	output/training_campaigns/2026-06-04-11-36-09_track2f_offset_aware_probe_campaign_2026_06_03/campaign_best_run.yaml
Execution report	output/training_campaigns/2026-06-04-11-36-09_track2f_offset_aware_probe_campaign_2026_06_03/campaign_execution_report.md
Planning report	doc/reports/campaign_plans/track_2/2026-06-03-17-25-37_track2f_offset_aware_probe_campaign_plan_report.md
Campaign technical document	doc/technical/2026-06/2026-06-03/2026-06-03-17-25-37_track2f_offset_aware_probe_campaign.md
Model technical document	doc/technical/2026-06/2026-06-03/2026-06-03-18-18-20_track2f_sequential_residual_offset_probe.md
Closeout technical document	doc/technical/2026-06/2026-06-04/2026-06-04-12-28-46_track2f_campaign_closeout.md

Execution Summary

Field	Value
Campaign name	track2f_offset_aware_probe_campaign_2026_06_03
Started at	2026-06-04T11:36:09
Finished at	2026-06-04T12:04:47
Completed runs	3
Failed runs	0
Tested model type	sequential_residual_offset_probe
Tested direction scopes	global , Fw , Bw
Runner-level scalar first entry	te_sequential_residual_offset_probe_remote_fw

Directional Branch Results

Surface	Role	Run	Family	Test MAE	Test RMSE	Val MAE	Params
global	bidirectional	te_sequential_residual_offset_probe_remote_global	sequential_residual_offset_probe	0.003537	0.004005	0.003783	92,802
Fw	forward only	te_sequential_residual_offset_probe_remote_fw	sequential_residual_offset_probe_fw	0.003385	0.003931	0.003380	92,802
Bw	backward only	te_sequential_residual_offset_probe_remote_bw	sequential_residual_offset_probe_bw	0.003638	0.004280	0.003840	92,802

These three rows are the closeout result. The Fw row is the scalar first entry in this small campaign, but it does not replace the required global or Bw branch candidates.

Execution Details

Surface	Run	Status	Duration	Return Code
global	te_sequential_residual_offset_probe_remote_global	completed	9m 22s	0
Fw	te_sequential_residual_offset_probe_remote_fw	completed	12m 09s	0
Bw	te_sequential_residual_offset_probe_remote_bw	completed	7m 07s	0

Scalar Leaderboard

Rank	Surface	Run	Family	Test MAE	Test RMSE	Val MAE	Params
1	Fw	te_sequential_residual_offset_probe_remote_fw	sequential_residual_offset_probe_fw	0.003385	0.003931	0.003380	92,802
2	global	te_sequential_residual_offset_probe_remote_global	sequential_residual_offset_probe	0.003537	0.004005	0.003783	92,802
3	Bw	te_sequential_residual_offset_probe_remote_bw	sequential_residual_offset_probe_bw	0.003638	0.004280	0.003840	92,802

Technical Interpretation

The sequential residual-offset probe is execution-valid and provides the first learned branch that directly targets the mean-offset failure mode found by the TE Curve Verification Pipeline mean-centered diagnostic, CVP 1.4 , and CVP 1.5 .

The scalar test metrics do not beat the current program-level scalar training winner, te_periodic_gru_sequence_remote_Bw , which remains at test MAE 0.002344 . This is expected for a narrow offset-aware probe: the campaign was designed to test whether the offset branch is feasible, not to promote a new global scalar winner from pointwise training alone.

Wave 3.1 is also a clean non-harmonic baseline. The sequential_residual_offset_probe architecture does not include explicit harmonic forcing, periodic sin / cos feature expansion, RCIM harmonic indices, or a structured harmonic branch. It should therefore not be judged as the harmonic shape-preserving intervention. Its value is to show how far a causal feedforward-plus-sequence residual structure can go when the future comparison introduces new curve indices, multi-head shape/offset training, or composite losses.

The important closeout result is therefore structural. The repository now has three trained sequential residual-offset candidates that can be evaluated in the official curve-first TE Curve Verification Pipeline surface:

- global : bidirectional/general candidate.
- Fw : forward-only candidate.
- Bw : backward-only candidate.

Registry Effects

Registry Scope	Current Family Best	Test MAE	Interpretation
sequential_residual_offset_probe	te_sequential_residual_offset_probe_remote_global	0.003537	New global Wave 3.1 branch candidate
sequential_residual_offset_probe_fw	te_sequential_residual_offset_probe_remote_fw	0.003385	New forward-only Wave 3.1 branch candidate
sequential_residual_offset_probe_bw	te_sequential_residual_offset_probe_remote_bw	0.003638	New backward-only Wave 3.1 branch candidate

The campaign runner refreshed the family registries for all three branch families and updated the program registry. The program-level scalar best did not move because the existing Wave 2.2 periodic GRU backward-only model is still stronger on scalar test_mae .

TE Curve Verification Pipeline Boundary

TE Curve Verification Pipeline was not run as part of this closeout. Under campaign governance, the official curve-first verification remains a separate operator-approved workflow after the final campaign-results report and PDF are complete.

The next verification package should evaluate all three Wave 3.1 branches in parallel, not only the scalar-first Fw row. The goal is to measure whether the offset-aware structure improves curve following and mean-offset behavior on the matching global , Fw , and Bw surfaces.

Closeout Decision

The campaign is complete and successful from an execution standpoint: all three runs completed, the leaderboard and best-run artifacts exist, and the three family registries were refreshed.

From a modeling standpoint, Wave 3.1 is a completed feasibility branch. It is not promoted over the current scalar program winner until a separate official TE Curve Verification Pipeline curve-first refresh confirms a real curve-level gain.

Recommended Follow-Up

1. Keep all three Wave 3.1 branch candidates: `global` , `Fw` , and `Bw` .
2. Do not collapse Wave 3.1 to the scalar-first `Fw` candidate.
3. Keep Wave 3.1 as the clean non-harmonic baseline for future comparisons against `harmonic-offset`, `new-index`, `multi-head`, and `composite-loss` models.
4. Prepare the optional TE curve-first verification refresh as the next operator-launched step.